Fiscal Year:	FY 2025	**Task Last Updated:**	FY 03/27/2025
PI Name:	Blaber, Elizabeth Ph.D.		
Project Title:	Understanding the Brain-Liver-Gut Axis during Spaceflight and Aging		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:	NIH, BARDA, FDA	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	(1) Cell Culture		
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PI Organization Type:	UNIVERSITY	**Phone:**	518-276-6548
Organization Name:	Rensselaer Polytechnic Institute		
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Zip Code:	12180	**Congressional District:**	20
Comments:	NOTE: Moved to Rensselaer Polytechnic Institute in fall 2019 from NASA Ames Research Center.		
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA015N. Extended Longevity of 3D Tissues and Microphysiological Systems for Modeling of Acute and Chronic Exposures to Stressors
Start Date:	02/02/2023	**End Date:**	01/31/2027
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA HQ
Contact Monitor:	Koniges, Ursula	**Contact Phone:**	202-256-8786
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Flight Program:			
Flight Assignment:	Per the Research Announcement for this solicitation: this effort is a collaboration between NASA; the National Institutes of Health (NIH); the Biomedical Advanced Research and Development Authority (BARDA); and the Food and Drug Administration (FDA).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Cekanaviciute, Egle Ph.D. (NASA Ames Research Center) Juran, Cassandra Marie Ph.D. (Blue Marble Space) Trapecar, Martin Ph.D. (Johns Hopkins School of Medicine) Hinshaw, Robert (NASA Ames Research Center/ORAU)		
Grant/Contract No.:	80ARC023CA007		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposal aims to interrogate the effects of spaceflight on cell and tissue dysfunction along the brain-liver-gut axis and to determine the role of these perturbations in accelerating tissue degeneration associated with aging. Spaceflight is known to affect multiple physiological systems, and multiple aging hallmarks, including stem and progenitor senescence, mitochondrial and immune dysfunction, and systemic inflammation, are present during aging. Our multi-disciplinary and multi-institutional team has specific expertise in organ systems, microphysiological systems (MPS) and with the brain-liver-gut axis multiorgan system (MOMPS), making us uniquely qualified to conduct the research proposed here. Our specific objective is to assess the role of radiation exposure on the induction of oxidative stress, chronic inflammation, cellular senescence, and senescence associated mitochondrial dysfunction (SAMD), and to determine the impact of these changes to the cellular and metabolic dysfunction associated with aging and accelerated aging during spaceflight. We hypothesize exposure to spaceflight factors results in increased oxidative stress and chronic inflammation leading to induction of senescence and mitochondrial dysfunction, and disruption of the brain-liver-gut axis, causing tissue damage and premature aging. This hypothesis will be tested with two aims. In Aim 1, we will address current limitations to the longevity of brain, liver, and gut MPSs using novel biomaterials and bioengineering principles in combination with embedded slow-release growth factors to mimic the mechanical environment and growth factor/small molecule gradients observed in vivo during development. Then, in Aim 2, we will examine the role of chronic inflammation, spaceflight and aging on the brain, liver,and gut using both individual MPSs of the brain, liver, and gut and an interconnected multiorgan MPS assessing crosstalk along the brain-liver-gut axis. We will specifically focus on cellular senescence and mitochondrial dysfunction in conjunction with dysregulation of lipid metabolism. Finally, we will assess the specific role of radiation induced inflammation on premature aging across multiple systems by challenging nonirradiated MOMPS with conditioned media and inflammatory cytokines from irradiated MOMPS. This research is highly relevant to multiple Federal Agencies, including NASA (NASA Space Biology and the Human Research Program), the NIH (NCATS and NIA), and the FDA. Therefore, successful completion of this proposal will be highly significant through the development of new tools to gain improved understanding of the role of oxidative stress and chronic inflammation on mitochondrial dysfunction, senescence induction, and accelerated aging on Earth and during spaceflight, and new tools adaptable for automated biological payloads to understand human multi-organ responses to the deep space environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Potential Impact: -- Long-term, multicellular, complex human tissue models for studying responses to chronic exposures to spaceflight stressors mimicking deep space exploration. -- New tools adaptable for automated biological payloads to understand human multi-organ responses to spaceflight stressors. -- New mechanistic insights into the role of oxidative stress and chronic inflammation on accelerated aging on Earth and during spaceflight in human models.
Task Progress:	Spaceflight impacts biological systems in profound ways: gravitational changes from microgravity to hypergravity, ionizing radiation, hostile environments, and isolation. These factors are known to cause significant impairments in multiple physiological systems, posing a major risk for long-term human space exploration. Short-duration spaceflight and simulated deep space radiation exposure leads to acute changes to physiological systems including bone and muscle loss, cardiovascular damage, activation of inflammatory pathways coupled with immune dysfunction, central nervous system (CNS) damage, and changes in stem cell regeneration. Although the obvious impacts of spaceflight are well documented, the underlying molecular mechanisms causing them are still being elucidated. In-depth investigations into immune system alterations, metabolic dysfunction, and cellular responses simultaneously in multiple organ systems in response to spaceflight are required to determine the cause of systemic alterations and to provide potential countermeasures. However, research progress and translation of spaceflight research to clinical studies has been limited by several factors, including lack of accessibility to human samples, size limitations of experimental platforms, crew time constraints during spaceflight exposure, and lack of automation. This proposal aims to interrogate the effects of spaceflight on cell and tissue dysfunction along the brain-liver-gut axis and to determine the role of these perturbations in accelerating tissue degeneration associated with aging. Our multi-disciplinary and multi-institutional team has specific expertise in organ systems, microphysiological systems (MPS), and with the brain-liver-gut axis multi-organ system (MOMPS), making us uniquely qualified to conduct the research proposed here. Our specific objective is to assess the role of radiation exposure on the induction of oxidative stress, chronic inflammation, cellular senescence, and senescence associated mitochondrial dysfunction (SAMD), and to determine the impact of these changes to the cellular and metabolic dysfunction associated with aging and accelerated aging during spaceflight. In Aim 1, we have addressed current limitations to the longevity of brain, liver, and gut MPSs using novel biomaterials and bioengineering principles to mimic the mechanical environment and growth factor/small molecule gradients observed in vivo during development. More specifically, we have designed individual growth chambers for each organ system resulting in extension of longevity of the organ systems for up to 6 months. Functional testing of the organ systems have demonstrated the growth environments are effective at maintaining both viability and functionality. Additionally, we have designed and implemented a multi-organ growth environment for intra-organ communication studies. Optimization of universal media that facilitates the growth of all three organ systems was essential to enabling an interconnected system. Thus, we have now accomplished the majority of our Aim 1 objectives. In Aim 2, we will examine the role of chronic inflammation, spaceflight, and aging on the brain, liver, and gut using both individual MPSs of the brain, liver, and gut and an interconnected multi-organ MPS assessing crosstalk along the brain-liver-gut axis. We will specifically focus on cellular senescence and mitochondrial dysfunction in conjunction with dysregulation of lipid metabolism.
Bibliography Type:	Description: (Last Updated: 04/03/2025)

Abstracts for Journals and Proceedings	Hinshaw R, Leyton T, Juran C, Blaber EA, Trapecar M, Costes S, Sowa M, Cekanaviciute E. "Automated maintenance of cerebral organoids on a microfluidics chip for a multi-organ microphysiological system." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024
Abstracts for Journals and Proceedings	Whitman B, Hinshaw R, Cekanaviciute E, Juran C, Trapecar M, Blaber EA. "Enhancing functional hepatocyte cells for developing a human brain-liver-gut microphysiological system." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024
Abstracts for Journals and Proceedings	Whitman B, Hinshaw R, Cekanaviciute E, Juran C, Trapecar M, Blaber EA. "Enhancing functional hepatocyte cells for developing a human brain-liver-gut microphysiological system." 2024 Annual Meeting of the Biomedical Engineering Society (BMES), Baltimore, MD, October 23-26, 2024. Abstracts, 2024 Annual Meeting of the Biomedical Engineering Society (BMES), Baltimore, MD, October 23-26, 2024. , Oct-2024
Abstracts for Journals and Proceedings	Whitman B, Hinshaw R, Cekanaviciute E, Juran C, Trapecar M, Blaber EA. "Understanding the impact of spaceflight on the brain-liver-gut axis: Using novel tools to adapt liver organoids to spaceflight conditions." The Microphysiological Systems (MPS) World Summit 2024, Seattle, WA, June 10-14, 2024. Abstracts. The Microphysiological Systems (MPS) World Summit 2024, Seattle, WA, June 10-14, 2024. , Jun-2024
Abstracts for Journals and Proceedings	Hinshaw RG, Leyton Reto TM, McFarland B, Zheng L, Juran C, Trapecar M, Blaber E, Costes SV, Sowa MB, Cekanaviciute E. "Developing long-lasting human cerebral organoids for integration into a multi-organ microphysiological system." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Leyton Reto TM, Hinshaw RG, Juran CM, McFarland B, Zheng L, Costes SV, Cekanaviciute E. "A microfluidic culture platform for long-term maintenance of cerebral organoids and integration into a multi-organ microphysiological system." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025